

CP Assignment –7

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1Q) Tea.Party.

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(NULL);

    int n,w;
    cin>>n>>w;

    vector<pair<int,int>>a(n);
    vector<int>ans(n);

    int sum=0;

    for (int i=0;i<n;i++) {
        cin >> a[i].first;
        a[i].second =i;

        ans[i] =(a[i].first+ 1)/2;
        sum += ans[i];
    }
    if (sum>w){
        cout<<-1<<"\n";
```

```

        return 0;
    }
    w -=sum;

    sort(a.rbegin(),a.rend());
    for (int i=0;i<n;i++) {
        int idx=a[i].second;
        int cap=a[i].first;

        int cad=cap-ans[idx];
        int ad=min(cad,w);

        ans[idx]+=ad;
        w -=ad;

        if (w==0)break;
    }

    for (int i=0;i<n;i++) {
        cout<<ans[i];
        if (i<n-1)cout<<" ";
    }

    cout<<"\n";

    return 0;
}

```

OUTPUT

Output

Status : Successfully executed

Time:

0.0000 secs

Memory:

3.6 Mb

Sample Input

2 10
8 7

Your Output

6 4

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 3.5 Mb

Sample Input

```
4 4
1 1 1 1
```

Your Output

```
1 1 1 1
```

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 3.432 Mb

Sample Input

```
3 10
9 8 10
```

Your Output

```
-1
```

2Q) Magic Powder-1.

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    ios::sync_with_stdio(false);
    cin.tie(NULL);
```

```

int T=1;
while(T--) {
    int n;
    long long k;
    cin>>n>>k;

    vector<long long> a(n), b(n);

    for (int i=0;i<n;i++)
        cin>>a[i];
    for (int i=0;i<n;i++)
        cin >> b[i];

    long long l=0,r=2000000005LL, ans=0;

    while (l<=r){
        long long mid =(l+r)/2;

        long long need =0;
        bool ok =true;

        for (int i=0;i<n;i++) {
            long long req = mid * a[i];
            if (req>b[i])need+=(req-b[i]);
            if (need>k) {
                ok=false;
                break;
            }
        }

        if (ok){
            ans=mid;
            l=mid+1;
        } else {
            r=mid-1;
        }
    }
    cout<<ans<<"\n";
}
return 0;
}

```

OUTPUT

Sample Input

3 1
2 1 4
11 3 16

Your Output

4

Output

Status : Successfully executed

Time:
0.0000 secs

Memory:
3.532 Mb

Sample Input

4 3
4 3 5 6
11 12 14 20

Your Output

3